

0 4

Mitochondrial DNA (mtDNA) is a small circular DNA molecule located in mitochondria. It is 16 569 nucleotides long and contains 37 genes and a control region.

Sports scientists investigated whether a mutation in the control region of mtDNA in human males was related to an ability to exercise for longer.

- The males in Group **T** had thymine at nucleotide position 16 519
- The males in Group **C** had a mutation resulting in cytosine at nucleotide position 16 519

0 4 . 1

The control regions of Group **T** and Group **C** were the same length.

Name the type of gene mutation that is most likely to have occurred at nucleotide position 16 519

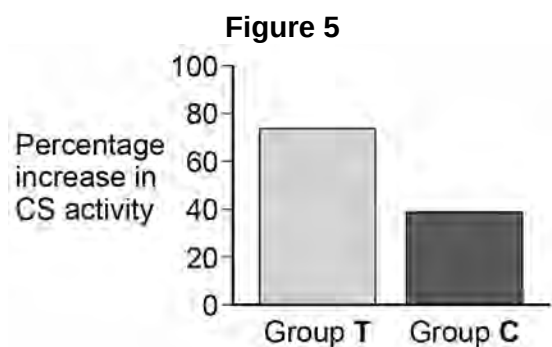
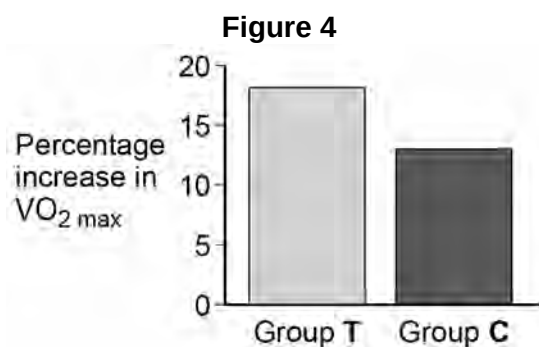
[1 mark]

Group **T** and Group **C** completed the same 8-week training programme. The following measurements were taken at the start of the 8-week programme, and again at the end.

1. $VO_{2\max}$ (a measure of maximal oxygen uptake).
2. Citrate synthase (CS) activity (CS is an enzyme involved in the Krebs cycle).

The scientists then calculated the percentage increase in each measurement in both groups.

Figure 4 and **Figure 5** show their results.



0 4 . 2

A student concluded from **Figure 4** and **Figure 5** that training has a positive effect on $\text{VO}_{2\text{max}}$ and CS activity.

Evaluate the student's conclusion.

[3 marks]

0 4 . 3

The mitochondrial DNA (mtDNA) control region is an area of mtDNA that is non-coding. This region stimulates the synthesis of both mtDNA and mitochondrial messenger RNA.

Use this information to suggest **two** reasons why the mutation at nucleotide position 16 519 could lead to the differences seen in **Figure 5**.

[2 marks]

1

2

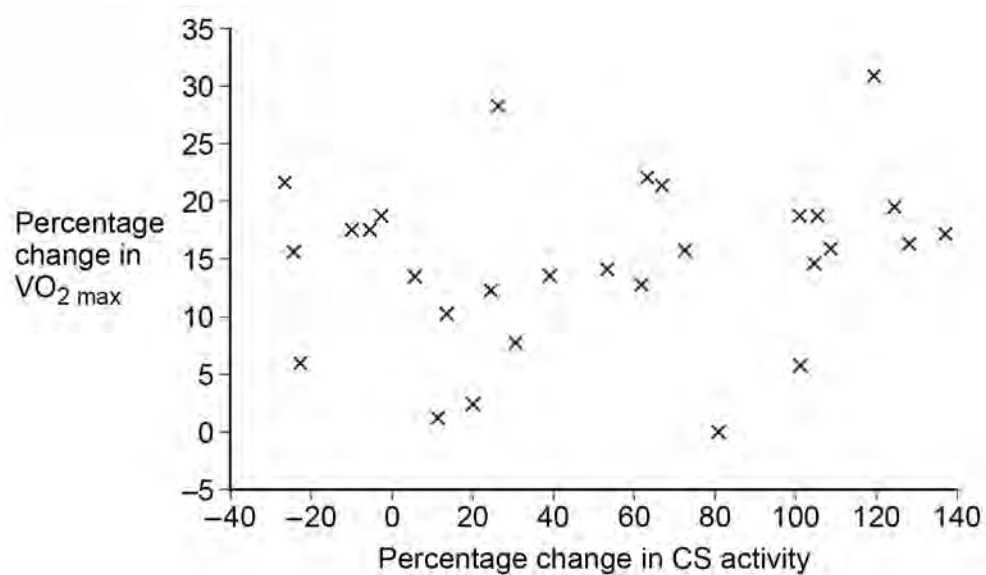
Question 4 continues on the next page

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The sports scientists investigated whether there was a correlation between the percentage change in $VO_{2\max}$ and percentage change in CS activity in Group T.

Figure 6 shows their results.

Figure 6



'Having thymine at nucleotide position 16 519 in Group **T** causes an increase in ability to exercise for longer.'

Use **all** the data in this question.

[illegible]

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